# CS & IT

## Compiler Design

DPP: 1

# Parsers

**Q1** Consider the following C program:

```
int main ( )
{
/*finding maximum element out of a & b*/
int a, b, max;
a = 10; b = 20;
if (a < b)
        max = b;
else
        max = a;
return (max);
}
```

Calculate the total number of tokens present in the program?

**Q2** Consider the following C-program:

```
1. int main( )
2. {
3. x = a + b* c;
4. y = x + a ;
5. char f = 'e' ;
6. in t g = 200;
7. ch/* comment ar = "gate";
8. }
```

Which of the following is/are correct regarding above program?

(A) The given program has 47 tokens.

(B) Given program produces compilation error

(C) Given program produces lexical error

(D) No error present in the program

**Q3** Compiler's first phase makes use of following patterns for token ($S_1$,$S_2$,$S_3$) recognition over the alphabet a,b,c.

$S_1$: b#(b|a)*c

$S_2$: c#(c|b)*a

$S_3$: a#(b|c)*b

**Note**: x# means 0 or 1 occurrence of the symbol x. The analyzer outputs the token that matched the longest possible prefix of the string. If abbbccccba is processed by first phase of compiler then which one of the following is the sequence of token of output.

(A) $S_1$, $S_2$, $S_3$ (B) $S_1$, $S_2$

(C) $S_3$, $S_2$ (D) $S_3$, $S_3$

**Q4** Consider the following C-program:

```
int main ( )
{
    int x; /* comment */
    x = = = = y /*abcd*/;
    int *p;
    int b = 10, y;
    x = *p +++++ y;
}
```

How many tokens are present in the given program?

**Q5** Consider the following code:

```
main ()
 {
     x = a/b + c;
     y = c – a;
     x = a + d;
     y = d/c + b;
}
```

Calculate the total number of token in the above code.

**Q6** Consider the following grammar G:

**G:** $S \rightarrow [S \mid SS \mid ]$

Where S is start variable, Number of terminals in follow set of S is_______

**Q7** Consider the following grammar.

$S \rightarrow (S) \mid SS \mid \in$

Total number of terminals in follow set of S is___.

**Q8** Consider the following Grammar G:

$S \rightarrow ACB$

$A \rightarrow a \mid b \mid \in$

$B \rightarrow c \mid d \mid \in$

$C \rightarrow e \mid f$

How many terminals are present in First (S)?

(A) 6 (B) 3

(C) 7 (D) 4

**Q9** Which of the following is/are correct regarding FIRST & FOLLOW of the given grammar.

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \in$

$T \rightarrow FT'$

$T' \rightarrow *FT' \mid \in$

$F \rightarrow id \mid (E)$

(A) FIRST (T)= {id, (}
FOLLOW(T') = {+, $, )}

(B) FIRST (E) = {id, (}
FOLLOW(E') = {$, )}

(C) FIRST (E) = {id, (}
FOLLOW(F) = {+, $ ,), *}

(D) FIRST (T') = {*, Î}
FOLLOW (E) = {$, * , )}

**Q10** Consider the following grammar G:

G:

$S \rightarrow AaAb \mid BbBa$

$A \rightarrow \in$

$B \rightarrow \in$

Number of terminals in First (S) is/are ____.

**Q11** Consider the following grammar:

$S \rightarrow P\,R\,Q \mid R\,b\,Q \mid Q\,a$

$P \rightarrow d\,a \mid Q\,R$

$Q \rightarrow g \mid \in$

$R \rightarrow h \mid \in$

The number of elements in Follow (R) and the follows set of R respectively is?

(A) 4, {$, b, g, h}

(B) 3, {b, g, h}

(C) 5, {$, a, b, g, h}

(D) 3, {$, b, g}

# Answer Key

| | |
|---|---|
| Q1 | 41~41 |
| Q2 | (B, C) |
| Q3 | (C) |
| Q4 | 34~34 |
| Q5 | 33~33 |
| Q6 | 3~3 |
| Q7 | 3~3 |
| Q8 | (D) |
| Q9 | (A, B, C) |
| Q10 | 2~2 |
| Q11 | (A) |

# Hints & Solutions

**Q1 Text Solution:**

The total tokens in the program are:

int main ( ) = 4

{ =1

/*finding maximum element out of a & b*/ = 0 (Because comment)

int a, b, max; = 7

a = 10; b = 20; = 8

if (a < b) = 6

max = b; = 4

else = 1

max = a; = 4

return (max); = 5

} = 1

Total token in above program are: 4 + 1 + 0 + 7 + 8 + 6 + 4 + 1 + 4 + 5 + 1 = 41

**Q2 Text Solution:**

The given program generates compilation and lexical error.

Therefore, option b, c are correct.

**Q3 Text Solution:**

Minimum string of $S_1$: c

Minimum string of $S_2$: a

Minimum string of $S_3$: b

It is given that prefer longest matching. So,

$$\underbrace{a\ b\ b\ b\ c\ c\ c\ b}_{S_3}\ \underbrace{a}_{S_2}$$

Therefore, option (c) is correct answer.

**Q4 Text Solution:**

The given program is:

$\underset{1}{\text{int}}\ \underset{2}{\text{main}}\ \underset{3}{(}\ \underset{4}{)}$

$\underset{5}{\{}$

$\underset{6}{\text{int}}\ \underset{7}{\text{x}}\ \underset{8}{;}$/*comment*/

$\underset{9}{\text{x}}\ \underset{10}{==}\ \underset{11}{=}\ \underset{12}{\text{y}}$/*abcd*/ $\underset{13}{;}$

$\underset{14}{\textit{int}}\ \underset{15}{*}\ \underset{16}{P}\ \underset{17}{;}$

$\underset{18}{\text{int}}\ \underset{19}{\text{b}}\ \underset{20}{=}\ \underset{21}{10}\ \underset{22}{,}\ \underset{23}{\text{y}}\ \underset{24}{;}$

$\underset{25}{\text{x}}\ \underset{26}{=}\ \underset{27}{*}\ \underset{28}{\text{p}}\ \underset{29}{++}\ \underset{30}{+}\ \underset{31}{++}\ \underset{32}{\text{y}}\ \underset{33}{;}$

$\underset{34}{\}}$

There are total 34 tokens in program.

**Q5 Text Solution:**

$\underset{1}{\text{main}}\ \underset{2}{(}\ \underset{3}{)}$

$\underset{4}{\{}$

$\underset{5}{\text{x}}\ \underset{6}{=}\ \underset{7}{\text{a}}\ \underset{8}{/}\ \underset{9}{\text{b}}\ \underset{10}{+}\ \underset{11}{\text{c}}\ \underset{12}{;}$

$\underset{13}{\text{y}}\ \underset{14}{=}\ \underset{15}{\text{c}}\ \underset{16}{-}\ \underset{17}{\text{a}}\ \underset{18}{;}$

$\underset{19}{\text{x}}\ \underset{20}{=}\ \underset{21}{\text{a}}\ \underset{22}{+}\ \underset{23}{\text{d}}\ \underset{24}{;}$

$\underset{25}{\text{y}}\ \underset{26}{=}\ \underset{27}{\text{d}}\ \underset{28}{/}\ \underset{29}{\text{c}}\ \underset{30}{+}\ \underset{31}{\text{b}}\ \underset{32}{;}$

$\underset{33}{\}}$

**Q6 Text Solution:**

Follow (s) = first (S)

= { [, ], $ }

Total 3 terminals in follow (S).

**Q7 Text Solution:**

$S \to (S) \mid SS \mid \varepsilon$

Follow (S) = {$, ), (}.

Number of non – terminals = 3

Hence, 3 is correct answer.

**Q8 Text Solution:**

First (S) = First (ACB)

First (S) = {a, b} U First (C)

= {a, b, e,f}

Number of terminals = 4

**Q9 Text Solution:**

Given that

| | **Frist** | **Follow** |
|---|---|---|
| E – | {(, id} | {$, )} |
| E' – | {+, Î} | {$ , )} |
| T – | {id, (} | {+, $ , ) } |
| T' – | {*, Î} | {+ , $ , )} |
| F – | {id, (} | {+, $, )} |

So, option a, b are correct.

**Q10 Text Solution:**

First (S) = {a,b}

Number of terminals = 2

**Q11 Text Solution:**

First (R) $\Rightarrow$ {h, $\varepsilon$}

Follow (R) $\Rightarrow$ First(Q) ∪ {b} ∪ Follow(P)

First (Q) $\Rightarrow$ {g, $\varepsilon$}

Follow (P) $\Rightarrow$ First (R) ∪ First (Q) ∪ Follow (S)

$\Rightarrow$ {h} ∪ {g} ∪ {$}

Follow (R) = {g} ∪ {b} ∪ {h, g, $}

= {b, h, g, $}.